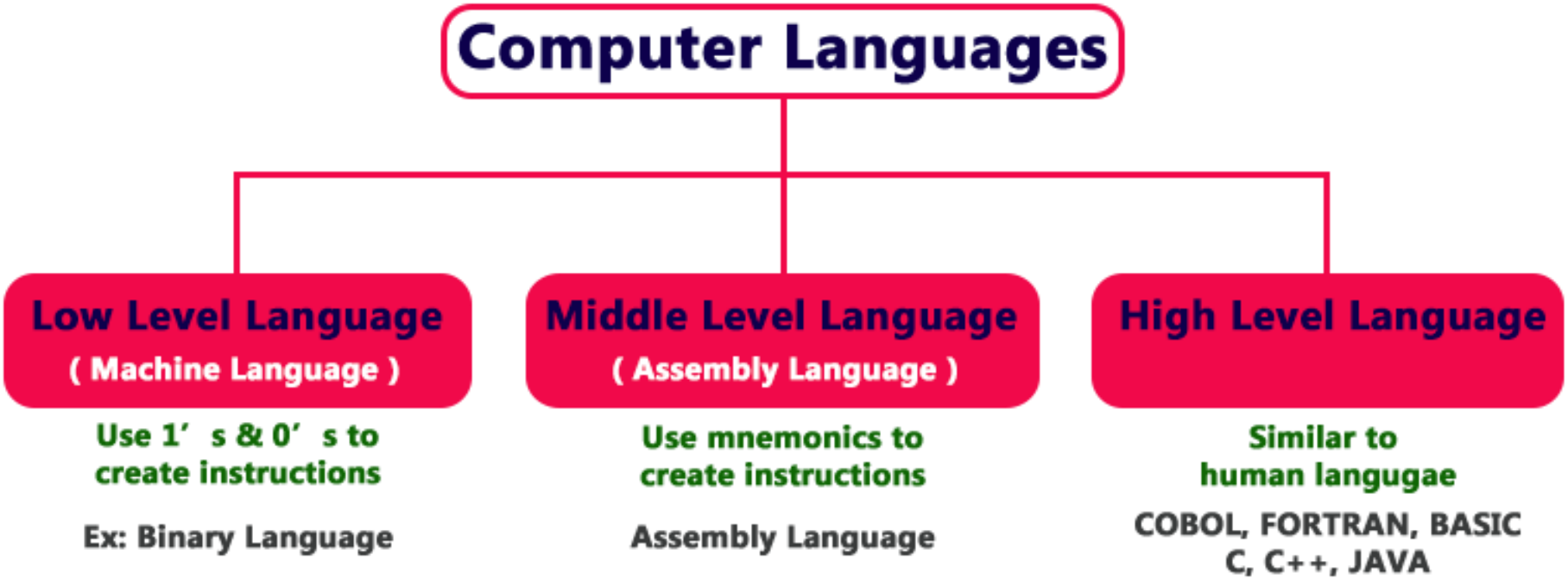
Introduction to Python Programming

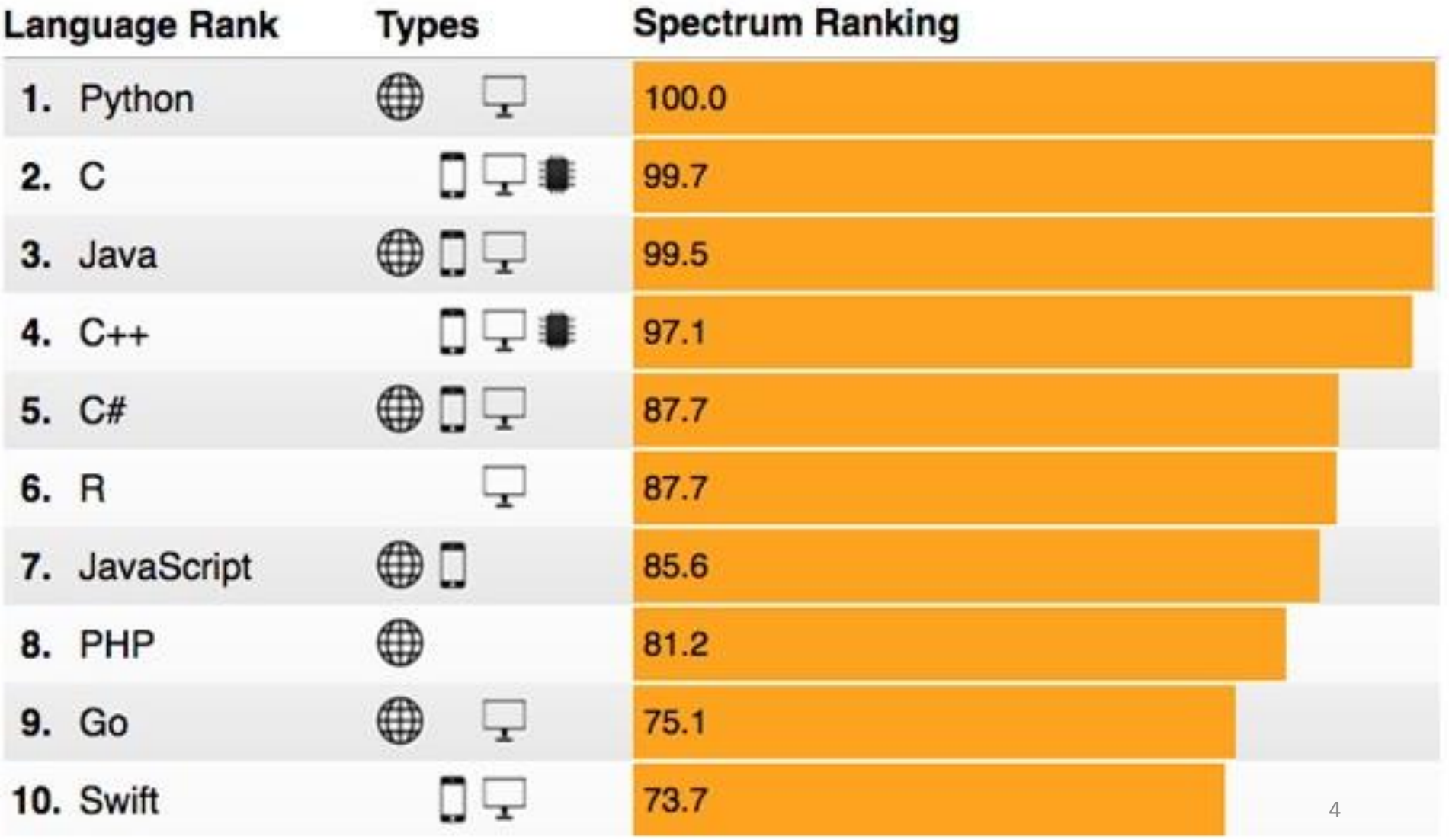
Elliot Attipoe

What is a programming language?

1. A programming language is a set of rules that provides a way of telling a computer what operations to perform.
2. A programming language is a vocabulary and set of grammatical rules for instructing a computer or computing device to perform specific tasks.
3. A programming language is a set of commands, instructions, and other syntax use to create a software program.


Computer Languages
Low Level Language
(Machine Language)
Use 1' s & 0' s to create instructions
Ex: Binary Language
Middle Level Language
(Assembly Language)
Use mnemonics to create instructions
Assembly Language
High Level Language
Similar to human langugae
COBOL, FORTRAN, BASIC
C, C++, JAVA

Top Ranking Programming Languages



About Python Programming Lang

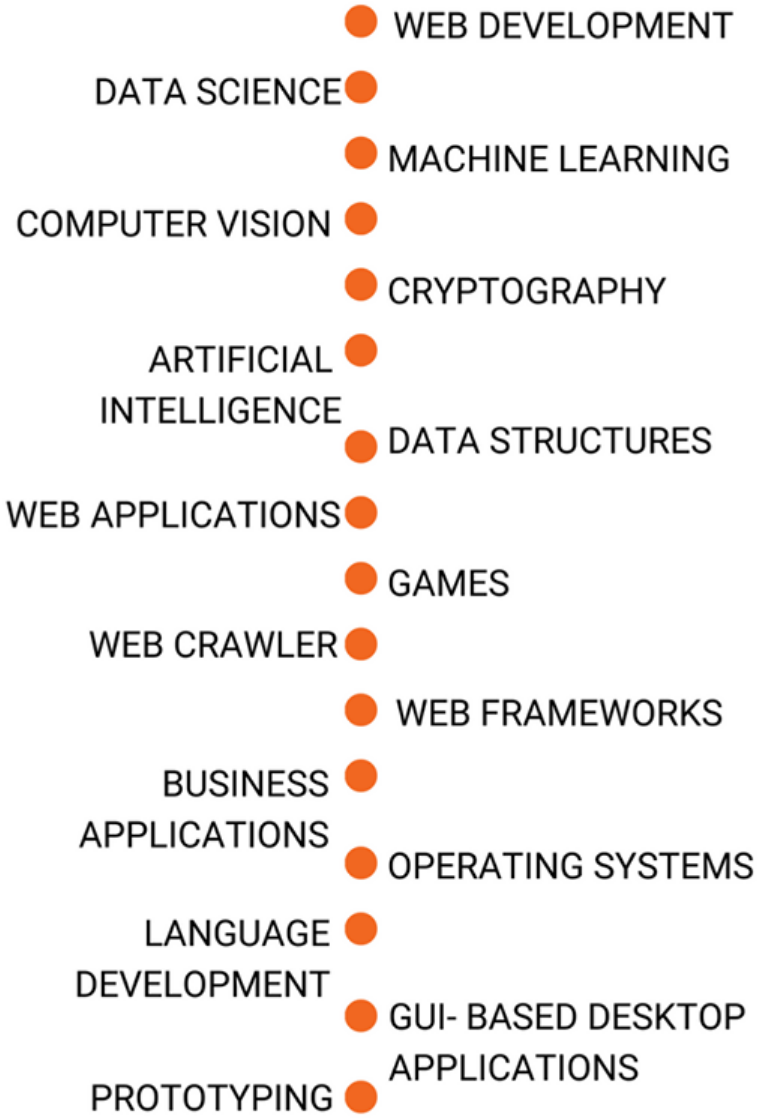
- Python is a widely used general-purpose, high level programming language.

- It was initially designed by **Guido van Rossum in 1991**

Python Applications



APPLICATIONS OF PYTHON



Top Companies using Python



Basic Mathematics Operations in Python

Operator	Name
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Modulus
**	Exponentiation
//	Floor/Integer Division

Evaluate the following expressions

1. `3*1**3`
2. `2 + 3 * 5 / 2`
3. `2 + 3 * 5 // 2`
4. `10 - 4 * 2`
5. `(10-4) * 2`
6. `6 + 3 - 5 * 4 / 2 % 2`

Augmented Assignment Operators

Operator	Name
+=	Addition assignment
-=	Subtraction assignment
*=	Multiplication assignment
/=	Float division assignment
//=	Integer division assignment
%=	Remainder assignment
**=	Exponent assignment

Assume i = 1, what are the results of the following expressions?

1. `i+=4`
2. `i-=4`
3. `i*=4`
4. `i/=4`
5. `i//=4`
6. `i%=4`

Example	*Equivalent*
`i += 8`	`i = i + 8`
`i -= 8`	`i = i - 8`
`i *= 8`	`i = i * 8`
`i /= 8`	`i = i / 8`
`i //= 8`	`i = i // 8`
`i %= 8`	`i = i % 8`
`i **= 8`	`i = i ** 8`

Comparison Operators

Operator	Operator Name
==	Equal to
!=	Not Equal to
>	Greater Than
<	Lesser Than
>=	Greater Than or Equal to
<=	Lesser Than or Equal to

What is the output of the following expressions

```
1.10 == 12

2.10 != 12

3.10 < 12

4.10 > 12

5.10 <= 12
```

Logical Operators

Operator	Operator Name
and	Logical AND
or	Logical OR
not	Logical NOT

Evaluate the following expressions

1. `True and False`
2. `True or False`
3. `not(True) and False`
4. `not(True and False)`
5. `(10 < 0) and (10 > 2)`
6. `(10 < 0) or (10 > 2)`
7. `not(10 < 0) or (10 > 2)`
8. `not(10 < 0 or 10 > 2)`

Boolean Logic Truth Table

P	Q	P and Q	P or Q	Not P
True	True	True	True	False
True	False	False	True	
False	True	False	True	True
False	False	False	False	

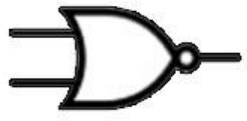
Basic Digital Logic Gates

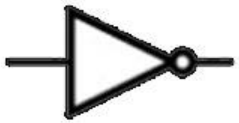
INPUT		OUTPUT
A	B	
0	0	0
1	0	0
0	1	0
1	1	1


AND

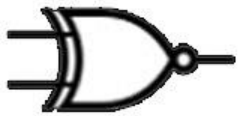

NAND

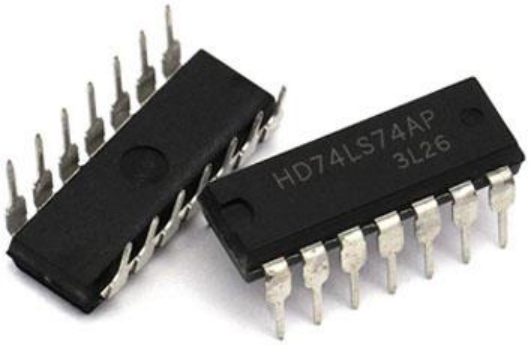

OR


NOR

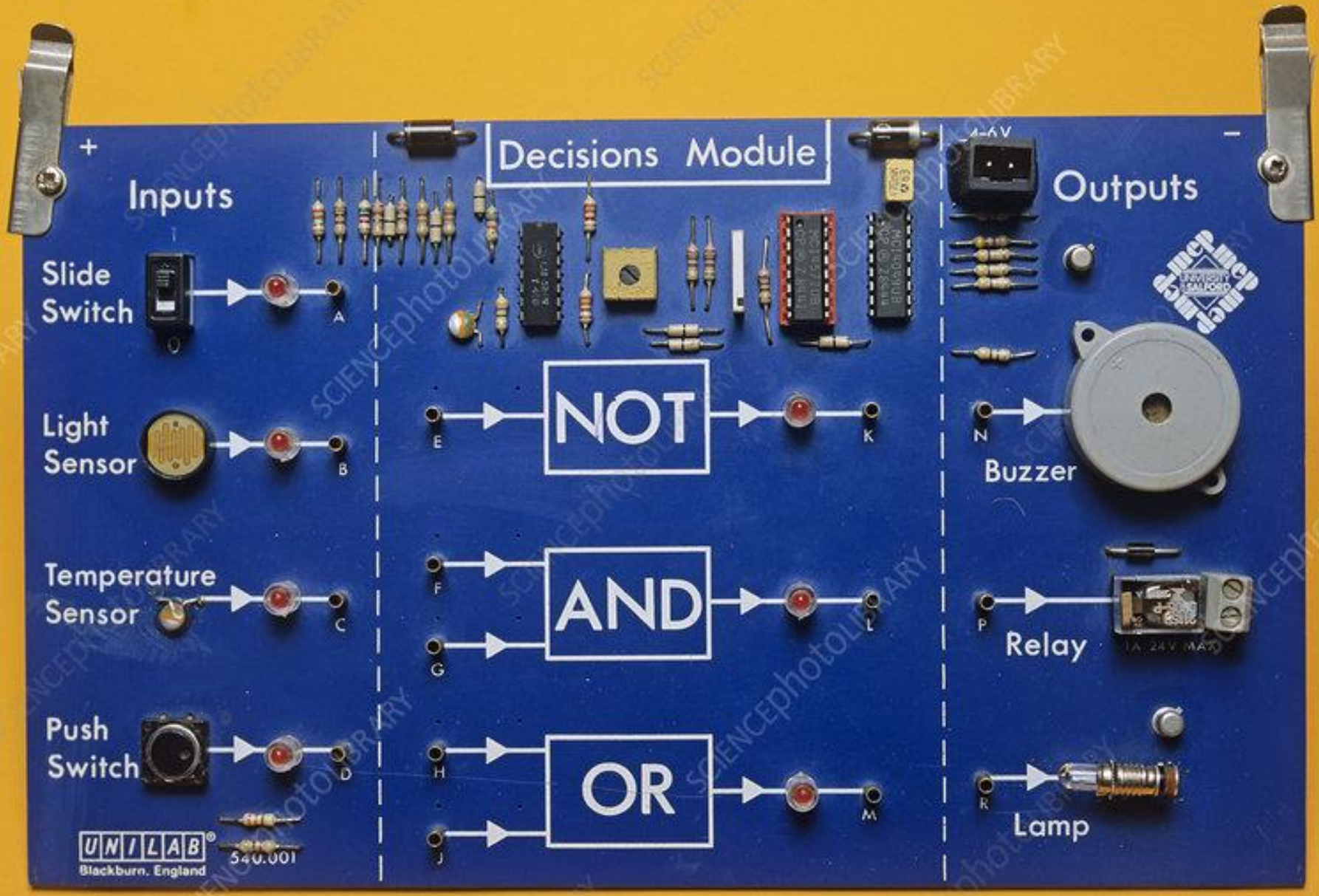

NOT

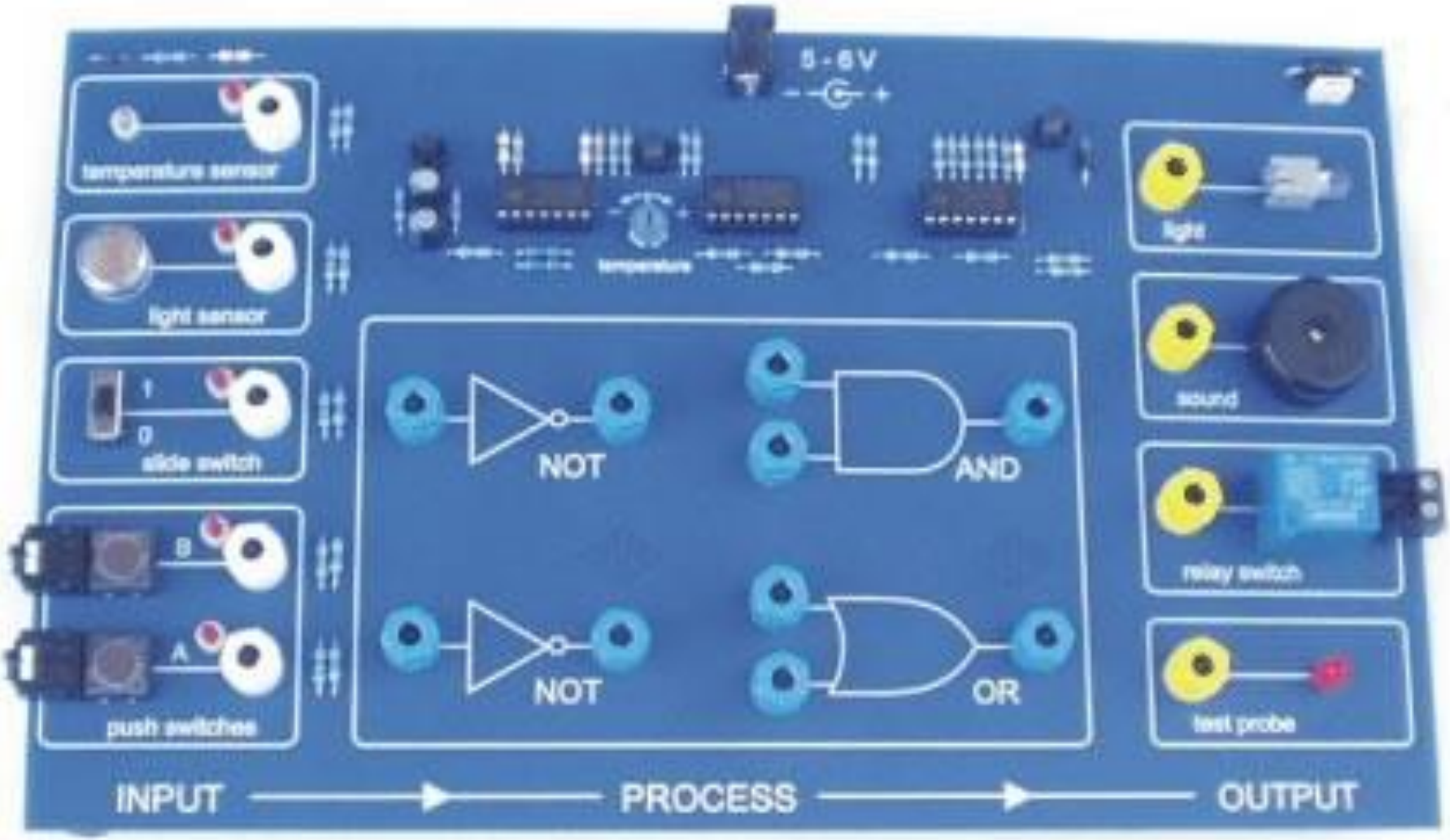

XOR

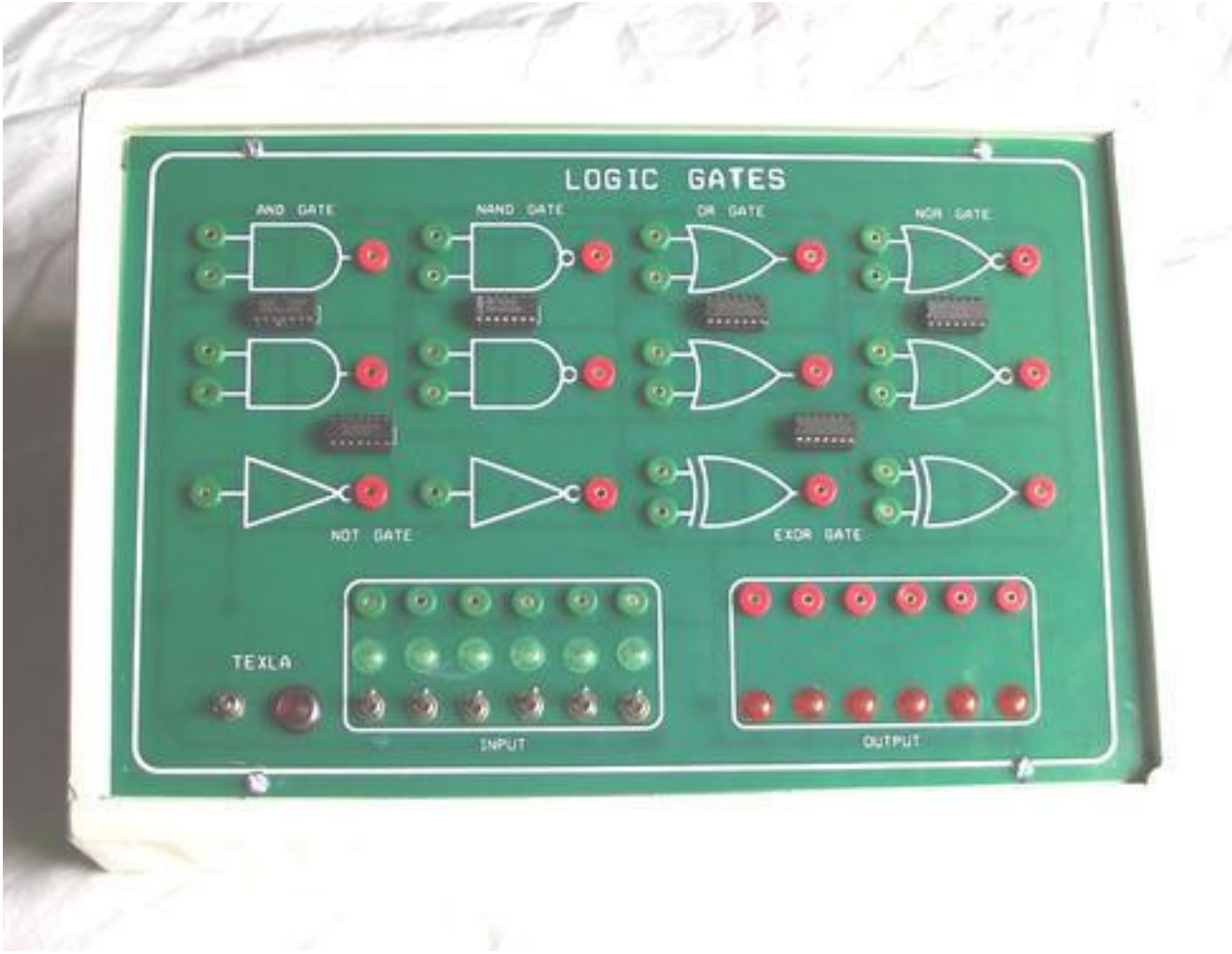

XNOR

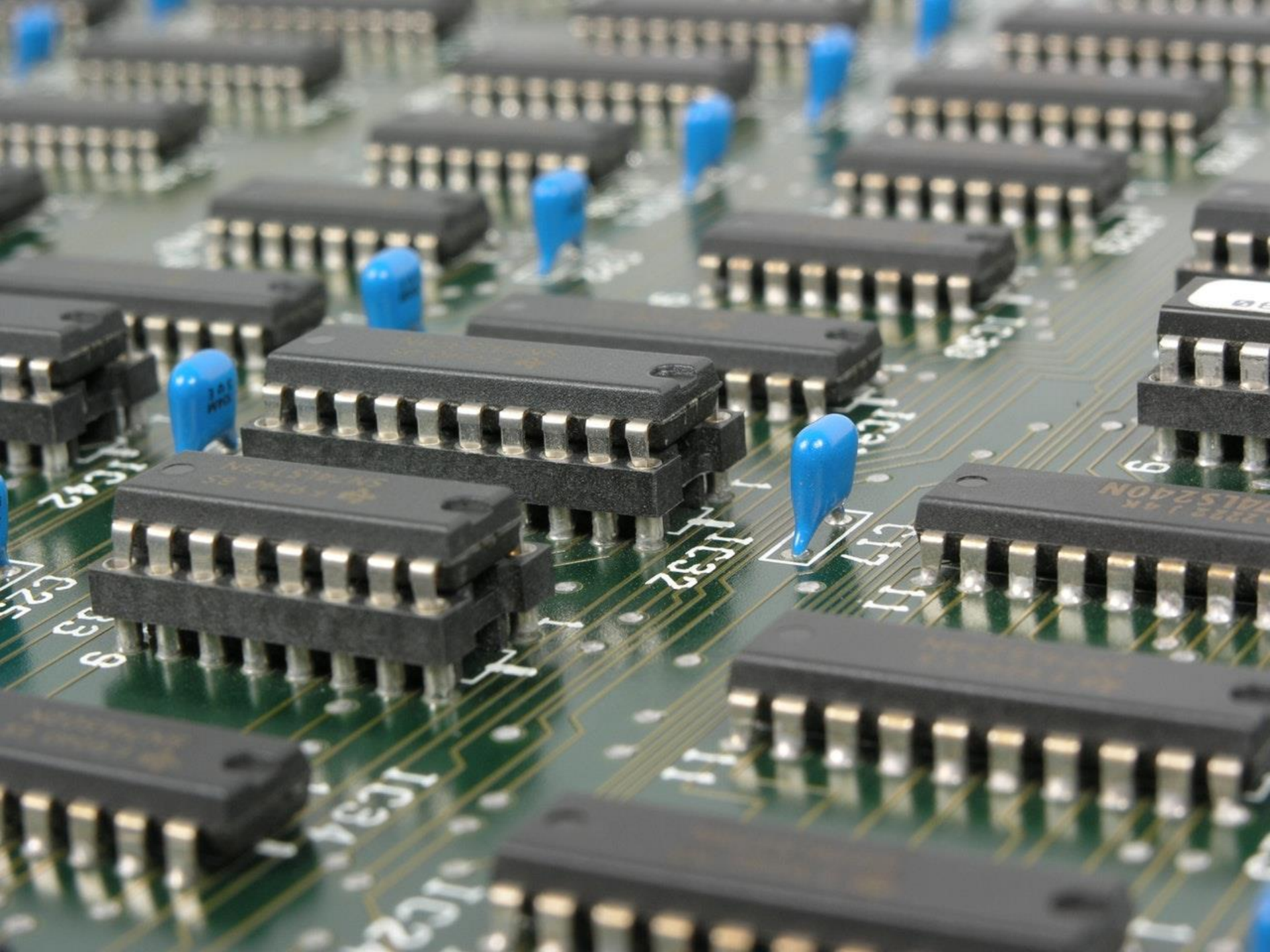


A AND B	$A \cdot B$
A OR B	$A + B$
NOT A	$\overline{A}$
A XOR B	$A \oplus B$


Decisions Module
Inputs
Outputs
+
−
Slide Switch
Light Sensor
Temperature Sensor
Push Switch
NOT
AND
OR
Buzzer
Relay
Lamp
A
B
C
D
E
F
G
H
J
K
L
M
N
P
R
UNILAB
Blackburn. England
540.001


5 - 6V
temperature sensor
light sensor
1
0
slide switch
B
A
push switches
NOT
AND
NOT
OR
light
sound
relay switch
test probe
INPUT
PROCESS
OUTPUT


LOGIC GATES
AND GATE
NAND GATE
OR GATE
NOR GATE
NOT GATE
EXOR GATE
TEXLA
INPUT
OUTPUT



Digital Logic Circuit

- Digital circuits make up all computers and computer systems.
- The operation of digital circuits is based on Boolean algebra Boolean Logic (AND, OR, and NOT)
- These basic functions can be combined in many ways to provide all the functions required in the central processor of a digital computer a digital computer.
- Digital circuits operate by performing Boolean operations on binary numbers

Precedence and Associativity

- Operator precedence determines the way in which operators are parsed with respect to each other.
- Associativity determines the way in which operators of the same precedence are parsed.
- Almost all the operators have left-to-right associativity.

Operator Precedence in Python

Operator	Meaning
( )	Parentheses
**	Exponent
*, /, //, %	Multiplication, Division, Floor division, Modulus
=, -	Addition, Subtraction
==, !=, >, >=, <, <=	Comparison
not	Logical NOT
and	Logical AND
or	Logical OR

Evaluate the following expression

a) `13 > 12 or 6 < 5`

b) `6 + 3 > 7 and 11 < 2 * 5`

c) `7 + 3 * 2 > 6 * 3 and True`

d) `5 * 4 > 6 ** 2 and True or False`

Variables

- A variable is a storage location (identified by a memory location) which contains some known or unknown quantity of information referred to as value.

- Programmers get to choose the names of the variables

Python Variable Name Rules

- Can start with a letter or underscore _
- Can consist of letters and numbers and underscore
- Is Case Sensitive

- **Good examples**
  - name, age tom25, _max
- **Bad examples**
  - 45name, #name, age.22
- **Different:** name, Name, NAME (case sensitive)

Reserved words in Python

```
Here is a list of the Python keywords.  Enter any keyword to get more help.

False               def                 if                  raise
None                del                 import              return
True                elif                in                  try
and                 else                is                  while
as                  except              lambda              with
assert              finally             nonlocal            yield
break               for                 not
class               from                or
continue            global              pass
```

Programming Errors

- Programming errors can be categorized into three types:
  - syntax errors
  - runtime errors
  - logic errors

Syntax Errors

- *Syntax errors* result from errors in code construction, such as mistyping a statement, incorrect indentation, omitting some necessary punctuation, or using an opening parenthesis without a corresponding closing parenthesis.

Runtime Errors

- *Runtime errors* are errors that cause a program to terminate abnormally.
- They occur while a program is running if the Python interpreter detects an operation that is impossible to carry out.
- Input mistakes typically cause runtime errors.

Logic Errors

- Logic errors occur when a program does not perform the way it was intended to.

- Errors of this kind occur for many different reasons.

Question

- Write a program to compute the volume of a sphere.

Assignment

- Write a program to compute the roots of a quadratic equation.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

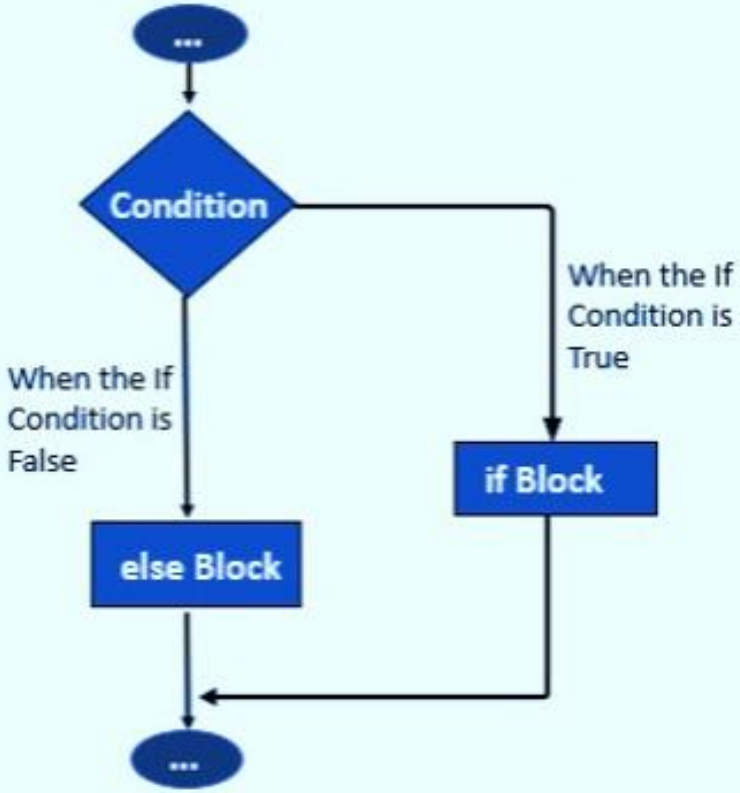
Submit source code via the e-learning platform

Due Date: May 2, 2022 @ 6pm

IF…Else statement

If Else in
Python





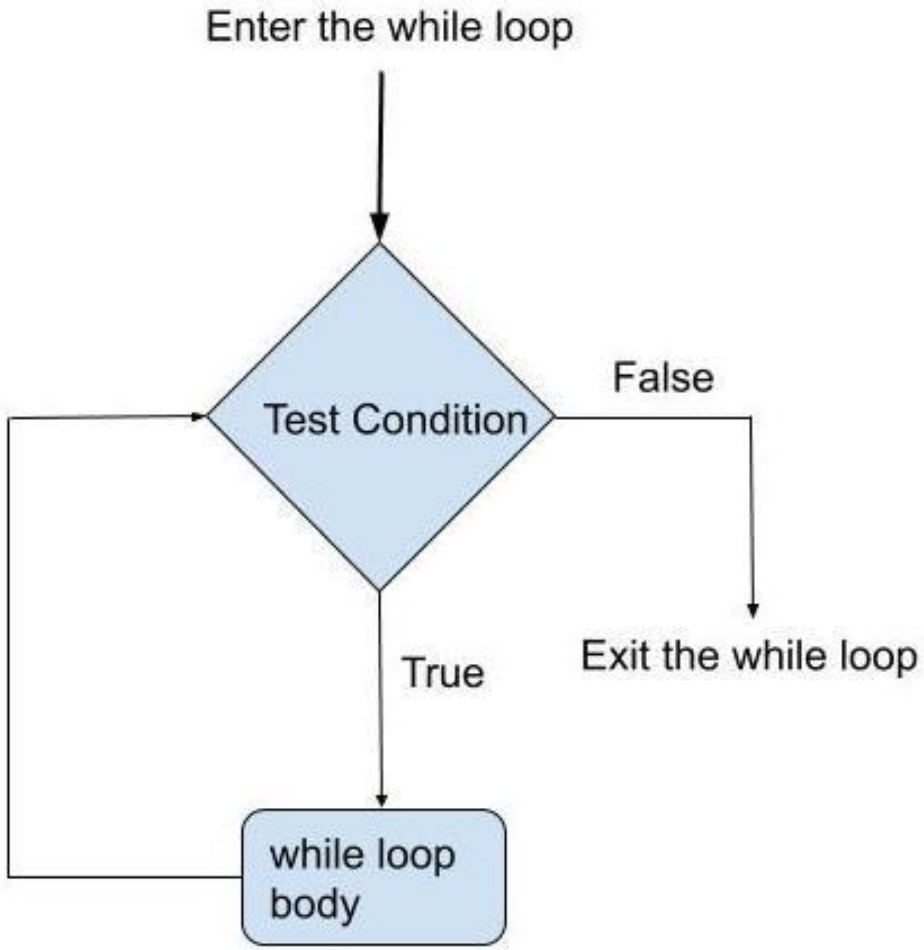
If...Else Example

exa.py - C:/Users/USER/Desktop/exa.py (3.10.2)

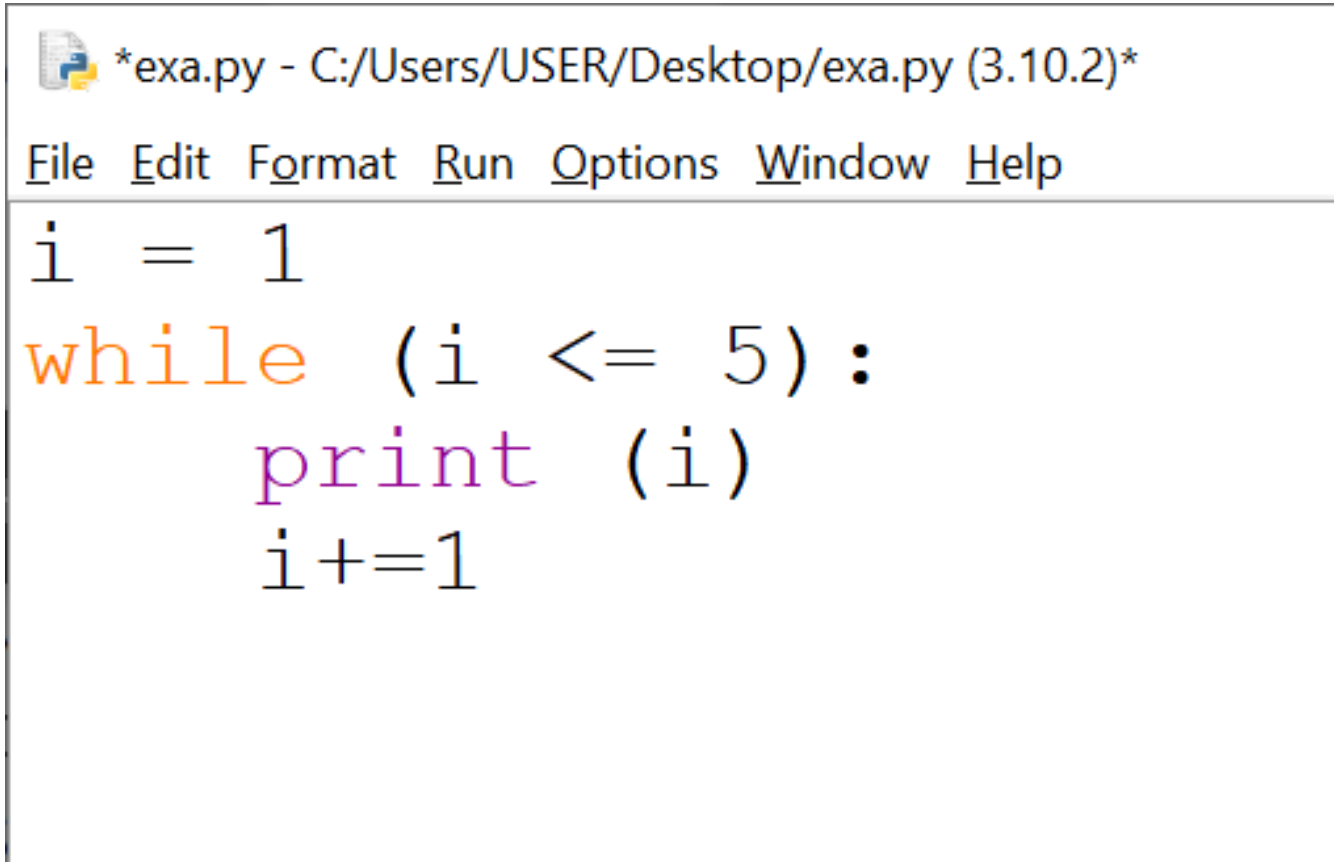
File Edit Format Run Options Window Help

```
num = eval(input("Enter a number : "))
if(num %2 == 0):
    print ("Number is Even")
else:
    print("Number is Odd")
```

While Loop


Enter the while loop
Test Condition
False
Exit the while loop
True
while loop
body

While Loop Example



For...Loop

```
1.for i in range (5):
2.    print ("Hello")
```

- `for i in range (5):`
- `    print (i+1)`

Assignment

- Use the turtle module in Python to draw the following:
  a) The Olympics rings logo
  b) The Ghana Flag

- Submit source code via the e-learning platform
- **Due Date:** May 2, 2022 @ 6pm